Density Worksheet

Questions

1. (a) Write the relationship between density ρ, mass m and volume V.
   (b) Rearrange the equation to make (i) m and (ii) V the subject.
   (c) State the SI units of density, mass and volume.
2. A sample has a mass of 2.5 kg and occupies a volume of $0.0025\,\text{m}^3$. Calculate its density.
3. A metal cube has a mass of 0.35 kg and a volume of $5.0 \times 10^{-5}\,\text{m}^3$. Calculate its density.
4. A liquid has a density of $800\,\text{kg/m}^3$. What volume does 4 kg of the liquid occupy?
5. Ice has a density of $920\,\text{kg/m}^3$. What is the mass of $0.015\,\text{m}^3$ of ice?
6. A block of material has a mass of 1.2 kg and a volume of $0.0013\,\text{m}^3$. Determine its density.
7. Aluminium has a density of $2700\,\text{kg/m}^3$. What is the mass of $0.004\,\text{m}^3$ of aluminium?
8. A sample with a mass of 0.60 kg has a density of $600\,\text{kg/m}^3$. Calculate its volume.
9. Gold has a density of $19\,300\,\text{kg/m}^3$. Find the mass of a piece of gold with a volume of $7.5 \times 10^{-4}\,\text{m}^3$.
10. A liquid sample has a mass of 15 g and a volume of $1.3 \times 10^{-5}\,\text{m}^3$. Calculate its density.
11. Cooking oil has a density of $900\,\text{kg/m}^3$. What volume is occupied by 2.7 kg of oil?
12. A gas occupies $0.10\,\text{m}^3$ and has a mass of 0.125 kg. Determine its density.
13. Steel has a density of $7850\,\text{kg/m}^3$. What volume does a 25 kg block of steel occupy?
14. Aluminium has a density of $2700\,\text{kg/m}^3$. Calculate the mass of $0.002\,\text{m}^3$ of aluminium.
15. A liquid sample has a mass of 3.5 kg and occupies $0.004\,\text{m}^3$. Find its density.
16. Water has a density of $1000\,\text{kg/m}^3$. Determine the volume that 0.5 kg of water will occupy.
17. **Extension.** A pine block measuring $0.20\,\text{m} \times 0.10\,\text{m} \times 0.05\,\text{m}$ has a mass of 750 g. Calculate its density in kg/m^3. (You must convert the mass to kilograms.)
18. A solid aluminium cylinder has a radius of 0.04 m and a height of 0.15 m. Given the density of aluminium is $2700\,\text{kg/m}^3$, determine its mass.
19. A cube of brass (density $8500\,\text{kg/m}^3$) has a mass of 1.7 kg. Calculate the length of one side of the cube.
20. A slab of marble measuring $0.50\,\text{m} \times 0.30\,\text{m} \times 0.020\,\text{m}$ is cut from a larger block. Find its mass, taking the density of marble as $2700\,\text{kg/m}^3$.
21. A solid steel sphere with radius 0.06 m is cast. Calculate its mass. (Density of steel: $7850\,\text{kg/m}^3$.)

Answer Key

1. (a) $\rho = \dfrac{m}{V}$

   (b) $m = \rho V$ $\qquad$ $V = \dfrac{m}{\rho}$

   (c) Density: $\mathrm{kg/m^3}$; Mass: kg; Volume: $\mathrm{m^3}$

2. $\rho = 1000\,\mathrm{kg/m^3}$
3. $\rho = 7.0 \times 10^3\,\mathrm{kg/m^3}$
4. $V = 5.0 \times 10^{-3}\,\mathrm{m^3}$
5. $m = 13.8\,\mathrm{kg}$
6. $\rho \approx 9.23 \times 10^2\,\mathrm{kg/m^3}$
7. $m = 10.8\,\mathrm{kg}$
8. $V = 1.0 \times 10^{-3}\,\mathrm{m^3}$
9. $m = 14.5\,\mathrm{kg}$
10. $\rho \approx 1.15 \times 10^3\,\mathrm{kg/m^3}$
11. $V = 3.0 \times 10^{-3}\,\mathrm{m^3}$
12. $\rho = 1.25\,\mathrm{kg/m^3}$
13. $V \approx 3.18 \times 10^{-3}\,\mathrm{m^3}$
14. $m = 5.4\,\mathrm{kg}$
15. $\rho = 8.75 \times 10^2\,\mathrm{kg/m^3}$
16. $V = 5.0 \times 10^{-4}\,\mathrm{m^3}$
17. $\rho = 7.5 \times 10^2\,\mathrm{kg/m^3}$
18. $m \approx 2.04\,\mathrm{kg}$
19. Side length $\approx 0.058\,\mathrm{m}$ (or $5.8\,\mathrm{cm}$)
20. $m = 8.10\,\mathrm{kg}$
21. $m \approx 7.10\,\mathrm{kg}$